

Guide to BigQuery

BigQuery is a powerful tool that cloud data analysts use to quickly analyze and understand large datasets, essential skills for making informed decisions in today's data-driven world. In this reading, you'll learn how to get started with BigQuery.

BigQuery

BigQuery is a cloud data warehouse designed to help data professionals manage and analyze data. It's serverless and scalable; BigQuery can also store and analyze your data stored in a variety of locations, including on-premises data centers, other cloud storage providers, and even streaming data.

As a cloud data analyst, BigQuery's data management and analysis tools make it simpler to handle your data workload. With BigQuery's serverless architecture, you don't need to manage or provision infrastructure on your own. Instead, BigQuery does it for you. And since it is scalable, you can add resources when you need it to handle spikes in the load, and scale down when the additional resources are no longer needed.

Access the BigQuery sandbox

BigQuery is part of the Google Cloud Platform. To manage your data needs, BigQuery is available in two tiers: Standard and Enterprise. To access either tier, you'll need to provide a credit card or set up a billing account and can incur charges for usage.

To access BigQuery:

1. Open a browser and navigate to the following URL in your browser:
<https://cloud.google.com/cloud-console>
2. Sign in to your account.
3. Locate BigQuery in the navigation menu.

To try out BigQuery's capabilities at no cost, you can use the BigQuery Sandbox. While some features in the sandbox are limited, the sandbox is a great way to get interactive experience with BigQuery's user interface and capabilities.

To enable the BigQuery sandbox:

1. Navigate to the following URL in your browser:
<https://console.cloud.google.com/bigquery>
2. To access the BigQuery Sandbox, you'll need to have a Google account. You can either sign in with your existing Google account or create a new one if you don't already have one.

3. On the welcome page, select your country and agree to the terms of service. Click 'Agree and Continue' to move on.

Welcome Cloud Data Student!

Create and manage your Google Cloud instances, disks, networks, and other resources in one place.



Cloud Data Student
studentclouddata@gmail.com

[SWITCH ACCOUNT](#)

Country

United States ▼

Terms of Service

☐ I agree to the [Google Cloud Platform Terms of Service](#), and the terms of service of [any applicable services and APIs](#).

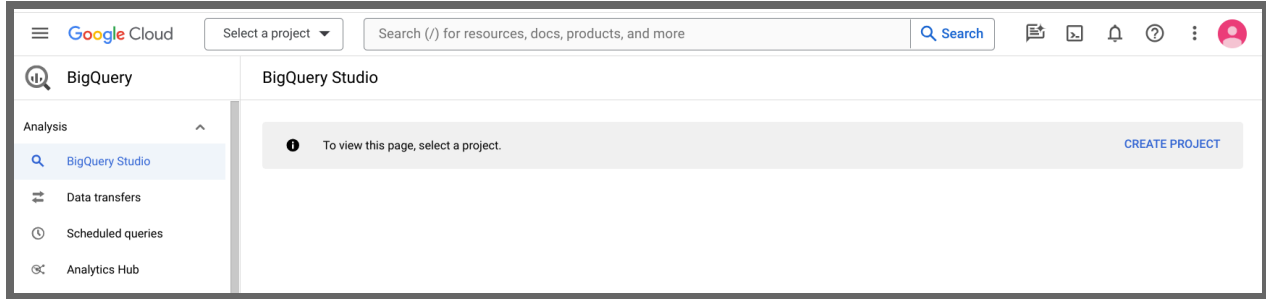
[AGREE AND CONTINUE](#)

Create a project

A project is a way to organize all of the resources you need in one place, including datasets, tables, views, and jobs. All datasets must belong to a project. In addition, projects are used to take costs and help keep your data secure. To work with BigQuery, you must create at least one project.

To create a project in BigQuery:

1. Click **Create Project**.



2. Give your project a meaningful name. Note: This name cannot be changed later!
3. The organization represents a group, such as a company. You do not need to select an organization to create a project. If you are not part of an organization, select **No Organization**.
4. Click **Create**.

New Project

 You have 12 projects remaining in your quota. Request an increase or delete projects. [Learn more](#)

[MANAGE QUOTAS](#)

Project name * 

Project ID: inspiring-orb-411518. It cannot be changed later. [EDIT](#)

Location * [BROWSE](#)

Parent organization or folder

[CREATE](#) [CANCEL](#)

Manage datasets

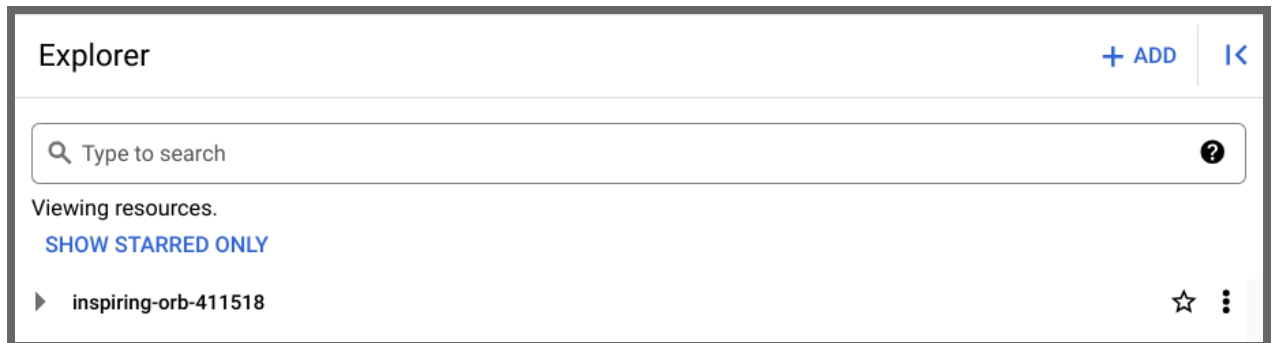
Datasets contain tables, views, and other data resources that you use for analysis and insights. Before using or adding a dataset, you must first select a Project.

To use an available dataset in a project:

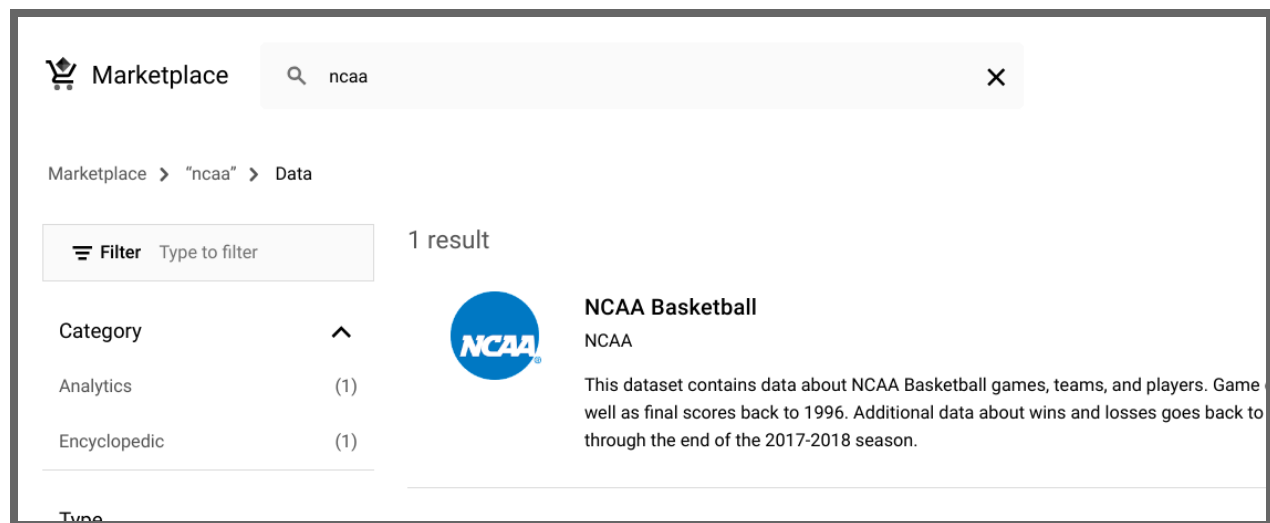
1. In the Explorer pane, scroll to see the available datasets.
2. Click on the arrow next to the dataset name to explore the datasets tables and other resources.

To add a public dataset to a project:

1. In the Explorer pane, click +Add.



2. In the search bar, enter public datasets, and then click the Public dataset icon.
3. In the Marketplace, search for the dataset by name. For example, enter NCAA in the search bar.
4. Select the dataset.
5. Click View Dataset to add the dataset to your project.



Query data

BigQuery supports both standard SQL and Google SQL to query data. To query data, you must first select a dataset.

1. In the Explorer pane, click on the arrow next to the dataset you want to query. This example uses the `ncaa_basketball` dataset.

Explorer

+ ADD | <

Viewing resources.

SHOW STARRED ONLY

▶	ga4_obfuscated_sample_ecommerce	☆	⋮
▶	gbif	☆	⋮
▶	gdelt_hathitrustbooks	☆	⋮
▼	ncaa_basketball	☆	⋮
	mascots	☆	⋮
	mbb_games_sr	☆	⋮
	mbb_historical_teams_games	☆	⋮
	mbb_historical_teams_seasons	☆	⋮
	mbb_historical_tournament_games	☆	⋮
	mbb_pbp_sr	☆	⋮

- Click on a table in the dataset, like the mascots table. This will open up the schema tab.
- In the schema table, click on a column, like the `mascots_name` column. This will open up a SQL editor.
- A query that returns the first 1000 values of that column will be displayed. Click Run Query to run the query and review the query results.

The screenshot shows the Google Cloud BigQuery console interface. At the top, there's a toolbar with buttons for RUN, SAVE, SHARE, SCHEDULE, and MORE. A status bar indicates 'Query completed'. The query editor contains the following SQL:

```
1 SELECT mascot_name FROM `bigquery-public-data.ncaa_basketball.mascots` LIMIT 1000
```

Below the query editor, the 'Query results' section is visible. It includes tabs for JOB INFORMATION, RESULTS (selected), CHART, PREVIEW, JSON, EXECUTION DETAILS, and EXECUTION. The RESULTS tab shows a table with 5 rows of mascot names:

Row	mascot_name
1	Captain Cane
2	Sparky
3	Mario the Magnificent
4	Kaboom!
5	Petey

5. Modify the existing query to get the results you need. For example, using the mascots table, you can modify the existing query to select only the mascots that have a name that starts with the letter 'A'.

Unset

```
SELECT mascot_name
FROM `bigquery-public-data.ncaa_basketball_mascots`
WHERE mascot_name LIKE `A%`;
```

6. Click Run again to see the modified results. Feel free to experiment with other queries and tables to practice querying in the BigQuery environment.

You can also access a new SQL editor window at any time by clicking Compose New Query +. This button is located in multiple places in the BigQuery UI to make it easy to query data when you need it.

Some places to find the Compose New Query+ button include:

- Top of the BigQuery homepage
- Datasets tab
- Query History tab
- Explore Query section

Key takeaways

BigQuery is a powerful data warehouse that can help you manage and analyze data on the cloud. To get started with BigQuery, access the Google Cloud platform with a Google account. If you don't have one, you can create one. Then, once logged into the Google Cloud Platform, you'll need to create at least one project to work with data in BigQuery. Within a project, you can work with and add datasets. In addition, the SQL editor allows you to run queries on datasets directly in the BigQuery interface. Big Query also offers a sandbox environment. This sandbox allows you to experiment with BigQuery's features without worrying about incurring usage charges.

Resources for more information

For more information, please visit the following resources.

- For more information on getting started with BigQuery sandbox, explore [Enable the BigQuery sandbox](#).
- For more information on public datasets and BigQuery, review [Query a public dataset with the Google Cloud Console](#).